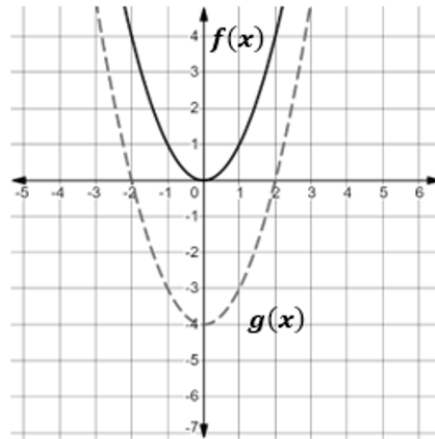


The graphs of  $y = f(x)$  and  $y = g(x)$  are shown.



1. Describe how to shift the graph of  $f(x)$  so that it coincides with the graph of  $g(x)$ .

Moves down 4 units

2. Use the graph to complete the table of values for  $f(x)$  and  $g(x)$ .

| $f(x)$ |     | $g(x)$ |     |
|--------|-----|--------|-----|
| $x$    | $y$ | $x$    | $y$ |
| -1     | 1   | -1     | -3  |
| 0      | 0   | 0      | -4  |
| 1      | 1   | 1      | -3  |

3. Explain the change in the  $y$ -values from  $f(x)$  to  $g(x)$ .

Subtract 4 from the  $y$ -values.

4. Complete the statement:

The graph of  $g(x)$  is a ☐ horizontal ☒ vertical shift of  $f(x)$ .

5. Complete the statement:

The shift is seen in the table by ☐ adding ☒ subtracting 4 to each ☐  $x$ -value ☒  $y$ -value

for each corresponding ☐  $x$ -value. ☒  $y$ -value.

For 6-7, complete the statements to describe how the graph of  $g(x) = f(x) + 7$  compares to the graph of  $f(x) = |x - 1|$ .

6. The graph of  $g(x)$  is a 

☐ horizontal

☒ vertical

 shift of  $f(x)$ .

7. The graph of  $g(x)$  is shifted 7 units 

☒ up from

☐ down from

☐ to the right of

☐ to the left of

 the graph of  $f(x)$ .

8. Describe the effect on the  $x$ -values of  $B(x) = |x + 5|$  for the transformation  $C(x) = B(x) - 7$ .

$C(x)$  is 7 below  $B(x)$ , the  $x$ -values are the same.

9. Describe the effect on the  $y$ -values of  $p(x) = \frac{4}{5}x + 2$  for the transformation  $r(x) = p(x) + 8$ .

$r(x)$  is 8 above  $p(x)$ , the  $y$ -values are all 8 more than the  $y$ -values of  $p(x)$ .

10. Describe how the graph of  $M(x) = N(x + 10)$  compares to the graph of  $N(x) = x^2 + 3$ .

$m(x)$  is 10 units to the left of  $n(x)$ .